

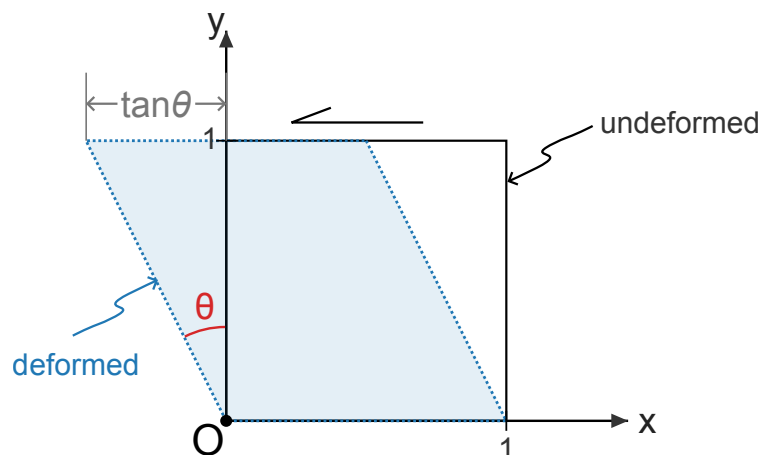
Finite Strain Analysis and Shear Zones

Instructions

1. No submission required for Questions 1 and 2. They will be discussed in class.
2. Provide brief answers to Questions 3 and 4. Also print out your **Jupyter Notebook** scripts associated with both question, (with all figure outputs included) for submission.
3. Submit your measurement steps and results for Question 4.

Question 1 (in class)

We have derived the $\mathbf{F}$ matrix for the pure shear deformation in class. Now derive the $\mathbf{F}$ for sinistral simple shear deformation with a shear strain of $\gamma = \tan \theta$. Show your reasoning. The following diagram of sinistral simple shear deformation is provided to you as a reference:



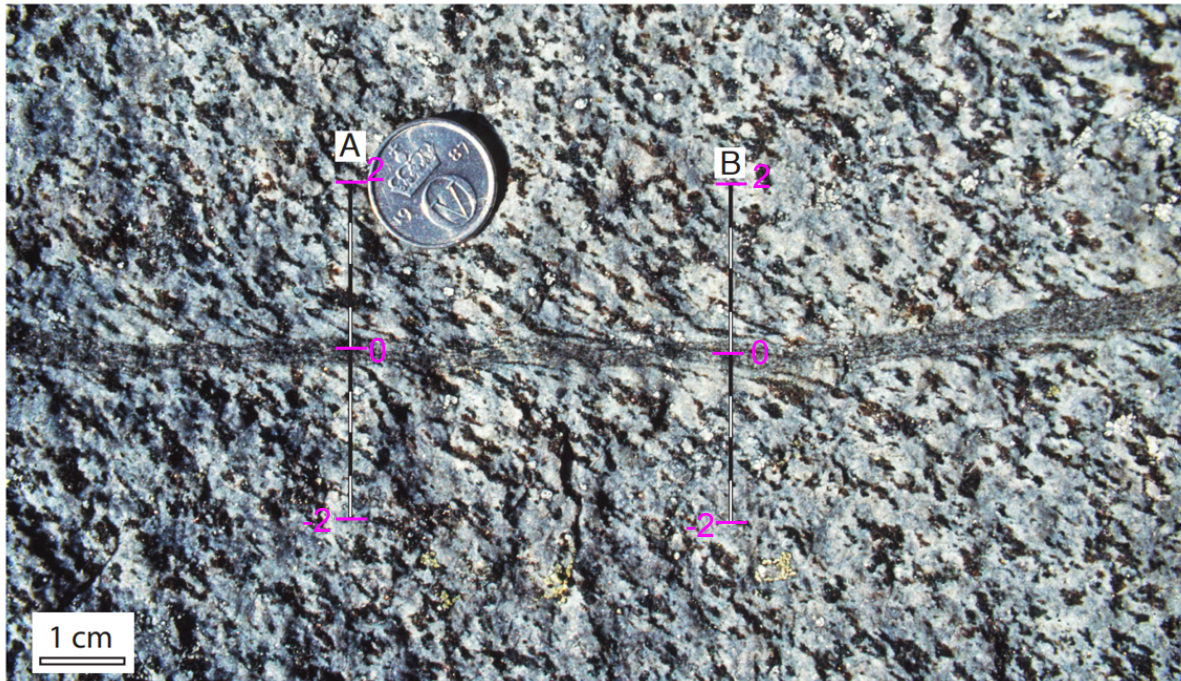
Question 2 (in class)

Rerun the `finite_strain_analysis.ipynb` with your derived $\mathbf{F}$ for the sinistral simple shear with increasing shear strain, from $\theta = 30^\circ$ to 60° . Show figure outputs from the **Jupyter** notebook and briefly explain:

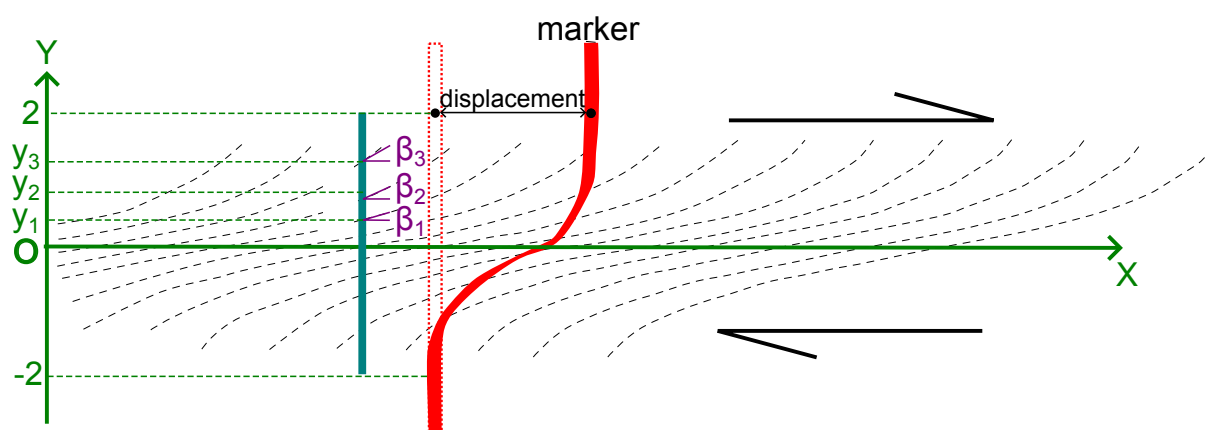
1. Why simple shear deformation is considered as *non-axial*?
2. What is the sense of rotation of the material lines? Are there any materials lines that are not rotated at all?
3. Are material lines experiencing shortened at $\theta = 30^\circ$ still under shortening as we accumulate shear strain and reach $\theta = 60^\circ$?

Question 3

The following figure displays a narrow shear zone developed in a magmatic rock. Now let's do some "forward modelling" on a virtual piece of rock sample. Plug the $\mathbf{F}$ you computed in Question 1 into `homogeneous_strain.ipynb`. Experiment with different shear strain (γ) values, do the modelled strain ellipsoids produce the similar deformation pattern as the natural shear zone in the figure?



Question 4



1. Create two shear strain profiles, A and B, across the shear zone. On the transparent paper, mark at least 8 oblique fabrics along each profile and measure the angles (β) between the foliation and the shear zone boundary. Record the position and β angle of the measurements.

2. Average the measurements along both profiles and input the values into the notebook file `heterogeneous_strain.ipynb` to construct a spatially varying shear strain γ profile. The β values has the following relationship with the shear strain γ : $\gamma = 2/\tan 2\beta$ and is already implemented for you in the notebook.
3. Compute the overall displacement d (cm) of the shear zone along each profile using the following relationship: $d = \int_{-2}^2 \gamma d\gamma$.
(Hint: consider using the function `scipy.integrate.simpson`)
4. Show the modelled deformation pattern from the figure. Is it more realistic than the previous model? What does that tell you about the strain distribution within the shear zone?